

Name: Tran, Thuc Bao · **Email:** thucbao.tran@sjsu.edu
Department: Computer Engineering, College of Engineering
Faculty Mentor: Prof. Kaikai Liu
Track: Undergraduate Research Project (\$1,500)
Period: Fall 2026 – Spring 2027 · Final report due 30 May 2027

Which Component Fails When a Roadside Detector Moves?

Synopsis of Proposed Research (240 words)

Roadside 3D detectors are trained at one intersection and deployed at another. Published accuracy is measured on the training distribution; deployment is not. The loss is known to be large — a reported 70–90% drop in detection rate across changes in lidar, geography and weather [1] — but the field reports it as a single number and attempts to close it wholesale.

Preliminary measurements on a Jetson Orin Nano suggest the loss is not uniform. Evaluating NVIDIA's pretrained V2XFusion [12] zero-shot from DAIR-V2X-I (Beijing) [10] to TUMTraf Intersection (Munich) [11], detection and position largely survive while heading estimation collapses: vehicles within $\pm 5^\circ$ of true heading fall from 88% to 28%, with the median error unchanged at -0.2° in both domains. Because roughly 20° of heading error on a 4.1×1.9 m box clears IoU 0.25 but not IoU 0.5, this one mode explains why Car accuracy falls $69.70 \rightarrow 0.13$ AP while Pedestrian and Cyclist retain 56% and 65%.

Heading is already treated as a separate error axis in-domain — nuScenes reports mAOE independently of mATE and mASE [2] — but a systematic search of the adaptation literature found it isolated only in-domain or under adversarial perturbation, never under natural cross-dataset shift, and never for roadside sensors.

This project will test whether domain fragility concentrates in specific components, and whether the fragile one can be repaired without target labels. Three label-free interventions will be measured against a supervised oracle.

A. Introduction

A.1 Current Research

The cross-dataset drop is large and label-free adaptation partly closes it. Self-training recovers 16–75% of the source-only-to-oracle gap across four LiDAR transfers using no target labels and no target statistics; on Waymo→KITTI it lifts Car AP_{3D} from 27.48 to 61.83 against a 73.45 oracle [3]. Cheaper still, a purely source-side augmentation — random object scaling — recovers about 86% of what the weakly supervised size prior buys (+27.19 AP_{3D} against +31.72), and *outperforms* it on cyclist [4].

Weak supervision is not automatically better. Statistical Normalization consumes target-domain object-size statistics yet degrades transfer when the size gap is small: nuScenes→KITTI AP_BEV falls $51.84 \rightarrow 40.03$, a -37.55% closed gap [3].

Orientation is already an established, separately-reported error axis. nuScenes defines mAOE as a standalone true-positive error alongside mATE and mASE, and computes mASE only after aligning orientation, decoupling scale from heading by construction [2]. Earlier work defines paired full- and half-range metrics, $\text{FOE} = |(\theta - \hat{\theta}) \bmod 360^\circ|$ and $\text{HOE} = |(\theta - \hat{\theta}) \bmod 180^\circ|$, so that 180° flips are distinguishable from angular imprecision [5].

Why heading is hard has a published explanation. Cui et al. observe that "the front and back of a vehicle may not be easily distinguishable from the LiDAR point cloud," and that parked vehicles fail because they "have no moving trajectory predictions that could be used to reliably infer the orientations" [5]. Heading is under-determined by appearance; detectors lean on motion to resolve it.

A.2 Limitations of Current Research

Every adaptation result above is vehicle-mounted and LiDAR-only. Waymo, KITTI, nuScenes and Lyft are all ego-vehicle datasets. A roadside sensor is static, has no ego-motion, and in the deployed single-frame configuration has no trajectory cue at all — precisely the cue detectors are shown to depend on for heading [5].

Heading has never been isolated under natural domain shift. It is measured in-domain (mAOE, FOE/HOE) and under adversarial perturbation, where error decomposition shows yaw is disproportionately sensitive and mAP-style metrics hide it [6]. MS3D++ treats heading as a class-dependent pseudo-label problem — catastrophic for

elongated vehicles because it destroys IoU, tolerable for BEV-symmetric pedestrians — but reports no yaw-specific metric [1]. None of this is cross-dataset.

Adaptation is evaluated as one number. No work reports which *component* failed, so it is unknown whether the loss is diffuse or concentrated, and therefore unknown whether a targeted, cheap repair is even possible.

A methodological trap is documented. Self-training's headline result is model-selection sensitive: Easy 3D AP has been reported fluctuating between 27.9% and 60.9% across randomizations, because best-epoch selection leaks target information [7]. A method that is label-free during training can become label-dependent at checkpoint selection.

A.3 Research Gap

A systematic, adversarially-verified literature search returned **no** cross-dataset results for roadside datasets (DAIR-V2X-I, Rope3D, TUMTraf), **no** LiDAR+camera fusion transfers, and **nothing** on INT8 quantization versus out-of-distribution robustness. That is a limit of the search rather than proof of absence — the roadside literature exists — but it establishes that per-component fragility for roadside 3D detection is, at minimum, not a well-covered question.

B. Specific Proposed Research and Why Important

Question. Does domain fragility in roadside 3D perception concentrate in specific network components, and can the fragile component be repaired using only unlabeled target data?

Preliminary evidence, already collected. A full pipeline was built and validated on a borrowed Jetson Orin Nano. It reproduces NVIDIA's published DAIR-V2X-I accuracy [10,12] to within 0.02–0.04 AP on Car and Pedestrian, establishing that subsequent measurements are trustworthy rather than artifacts. Zero-shot transfer to TUMTraf Intersection [11] (2,160 frames) gives:

3D AP, moderate	DAIR-V2X-I	TUMTraf	retained
Car @ IoU 0.5	69.70	0.13	0.2%
Pedestrian @ IoU 0.25	49.39	27.61	55.9%
Cyclist @ IoU 0.25	57.90	37.76	65.2%

vehicle heading error	DAIR-V2X-I	TUMTraf
median (signed)	−0.2°	−0.2°
standard deviation	12.4°	24.0°
within ±5°	88%	28%

The median is identical across domains, which excludes a coordinate or calibration error: the bias is correct and the reliability is not. The error is multi-modal, clustering near +15...30° and −30...−60° — consistent with a learned prior over road directions rather than random degradation. The class pattern independently matches MS3D++'s observation that heading error destroys IoU for elongated vehicles while symmetric classes tolerate it [1].

Why it matters. A missed detection is a failure mode downstream planners are built to tolerate. A confident *wrong heading* is not: heading feeds motion prediction, so a correctly located vehicle with a 25° heading error yields a predicted trajectory going somewhere the vehicle is not. If fragility is concentrated, repair can be local and label-free; if diffuse, retraining on labeled target data is the only option, which does not scale to per-intersection deployment.

C. Methodology and Why Innovative

C.1 Per-component fragility profile. Adapt the sub-task attribution protocol of monodle [8]: substitute ground truth for one predicted quantity at a time — heading, center, size, class — and re-measure AP. Applied both in-domain and cross-domain, the *difference* in each substitution's effect attributes the drop to specific components. monodle applies this in-domain to a monocular detector; applying it across a natural domain gap, to a LiDAR+camera roadside detector, is what is new.

C.2 Test the road-geometry hypothesis. If heading rests on a learned road-angle prior, the error clusters must align with the *source* intersection's lane bearings. This is directly falsifiable, and is scheduled first because a null result redirects the work.

C.3 Four label-free repairs, measured against an oracle.

intervention	supervision	precedent
Recalibrate INT8 ranges on unlabeled target frames	none — forward passes	untested in the verified literature
Adapt orientation head only, backbone frozen	pseudo-labels	—
Random object scaling at source pre-training	none — source-side	+27.19 AP _{3D} on car [4]
Local-structure input features	none — source-side	>21 mAP Waymo→KITTI [9]

A supervised oracle establishes the recoverable ceiling so each repair is reported as *fraction of gap closed*, the convention used by the self-training literature [3].

C.4 Guard against the documented traps. Checkpoints will be selected on a source-domain validation split, never on target performance, because best-epoch selection on the target leaks label information and inflates results [7]. Statistical Normalization will be included as a *negative control* rather than a baseline to beat, since it is known to degrade transfer when the size gap is small [3]. Per-class AP will be reported throughout, never a single averaged mAP, because the class balance differs between the two domains.

C.5 Compression as an axis, not the headline. Each repair is evaluated at FP16 and INT8. Preliminary data shows quantization adds 0.0° of heading error in-domain and +1.2° out-of-domain; establishing whether that is real requires repeated calibration seeds and per-class confidence intervals, which C.3 supplies.

Why innovative. Existing adaptation methods treat the detector as one object and ask how much AP returns. This asks *which part broke*, then repairs that part. It also moves the orientation question out of the vehicle-mounted, multi-frame setting where it was identified and into the static, single-frame roadside setting where the motion cue that previously rescued it does not exist.

D. Milestones and Timeline

period	milestone	deliverable
Sep–Oct 2026	Lane-bearing correlation (C.2); second target domain converted	Hypothesis confirmed or refuted
Nov–Dec 2026	Per-component fragility profile (C.1), two dataset pairs	Fragility table; figure set
Jan–Feb 2027	Supervised oracle; repairs 1–2 with repeated seeds	Fraction-of-gap-closed per repair
Mar 2027	Repairs 3–4; full FP16/INT8 matrix; Jetson latency for deployable repairs	Complete results
Apr 2027	Writing; SJSU Research Showcase	Draft manuscript
May 2027	Final report	Submitted by 30 May 2027

Risk and contingency. The oracle is the only step needing substantial GPU time; without it, results are reported as absolute recovery rather than fraction-of-oracle — weaker but publishable. If C.2 refutes the road-geometry hypothesis, C.1 and C.3 are unaffected.

E. Anticipated Outcome

1. **A per-component fragility profile** for roadside 3D detection across two dataset pairs — attributing a cross-dataset drop to specific components rather than a single AP number.
2. **A measurement of how much of the gap each label-free repair recovers**, with confidence intervals, comparable to the fraction-of-gap convention in the adaptation literature.
3. **The first cross-dataset heading-error measurement for roadside sensors**, reported as a metric distinct from AP.
4. **An answer on quantization:** whether INT8 amplifies domain fragility, measured on a mechanism-level metric sensitive enough to resolve it.
5. **Released artifacts** — the TUMTraf→DAIR converter, the JetPack 7 deployment recipe and all evaluation code, already public at github.com/henrytran412/God-eye.

Target venues: SJSU Research Showcase; a workshop paper at CVPR, ICRA or IROS.

Budget (\$300). Portable SSD for dataset storage (~\$120); Jetson power-monitoring hardware and cabling (~\$80); remainder for dataset access and incidentals.

References

1. Tsai, D., Berrio, J.S., Shan, M., Nebot, E., Worrall, S. *MS3D++: Ensemble of Experts for Multi-Source Unsupervised Domain Adaptation in 3D Object Detection*. IEEE T-IV, 2024. arXiv:2308.05988.
2. nuScenes detection benchmark — mAOE, mATE, mASE definitions. nuScenes devkit. github.com/nutonomy/nuscenes-devkit
3. Yang, J., Shi, S., Wang, Z., Li, H., Qi, X. *ST3D: Self-training for Unsupervised Domain Adaptation on 3D Object Detection*. CVPR 2021. arXiv:2103.05346.
4. Yang, J., Shi, S., Wang, Z., Li, H., Qi, X. *ST3D++: Denoised Self-training for Unsupervised Domain Adaptation on 3D Object Detection*. IEEE T-PAMI 2022. arXiv:2108.06682.
5. Cui, H., Chou, F.-C., Charland, J., Vallespi-Gonzalez, C., Djuric, N. *Uncertainty-Aware Vehicle Orientation Estimation for Joint Detection-Prediction Models*. 2020. arXiv:2011.03114.
6. Chandorkar, A. et al. *Comprehensive Robustness Analysis of LiDAR-based 3D Object Detection in Autonomous Driving*. 2026. arXiv:2607.02074.
7. Independent reproducibility analysis of ST3D self-training. 2024. arXiv:2408.12708.
8. Ma, X., Zhang, Y., Xu, D., Zhou, D., Yi, S., Li, H., Ouyang, W. *Delving into Localization Errors for Monocular 3D Object Detection*. CVPR 2021. arXiv:2103.16237.
9. Malić, D., Fruhwirth-Reisinger, C., Schuster, S., Possegger, H. *GBlobs: Explicit Local Structure via Gaussian Blobs for Improved Cross-Domain LiDAR-based 3D Object Detection*. CVPR 2025. arXiv:2503.08639.
10. Yu, H. et al. *DAIR-V2X: A Large-Scale Dataset for Vehicle-Infrastructure Cooperative 3D Object Detection*. CVPR 2022.
11. Zimmer, W. et al. *TUMTraF Intersection Dataset: All You Need for Urban 3D Camera-LiDAR Roadside Perception*. IEEE ITSC 2023.
12. NVIDIA. *CUDA-BEV Fusion and CUDA-V2X Fusion*, Lidar_AI_Solution. github.com/NVIDIA-AI-IOT/Lidar_AI_Solution